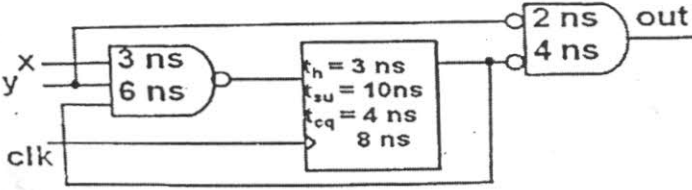


Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

Unit – I		Marks	BT*	CO*	PO*
1. a)	With the help of neat design flow, discuss SOC verification methodology.	10	L*2	1	1
b)	Explain the following terms in block level verification	10	L2	2	1
i.	Black box testing				
ii.	White box testing				
2. a)	What are IP blocks? Explain different types of IP blocks available for functional verification.	10	L2	1	1
b)	What is testbench creation? Briefly describe the different test bench creation techniques.	10	L2	2	1
3. a)	Compare VLSI design verification and Testing.	10	L2	1	1
b)	Discuss in detail system testbench metrics.	10	L2	2	1
Unit – II					
4. a)	For the circuit shown in Fig. Q4(a), calculate				
i.	Setup Time				
ii.	Hold time				
iii.	Minimum clock period				
iv.	Maximum clock frequency				
		10	L3	3	1
b)	What is crosstalk? What are the reasons for crosstalk? How does it create reasons for signal integrity?	10	L2	4	1
5. a)	With an example discuss on clock skew and short path problem. How can short path problem be eliminated?	10	L2	3	1
b)	Discuss in detail PVT variation and analysis corner.	10	L2	4	1
6. a)	Define				
i.	Setup time	10	L2	3	1
ii.	Hold time				
iii.	Propagation delay				
iv.	Contamination delay				
v.	Aperture time	10	L2	4	1
b)	Explain the signal integrity analysis steps with neat diagram.	10	L2	4	1
Unit – III					
7. a)	What is BIST process? With the neat diagram explain the basic architecture of BIST.	10	L2	5	1
b)	Briefly discuss the factors to be considered for the test plan. Based on which factors automatic test equipment is bought?	10	L2	5	1
8. a)	Explain the scan design process and scan design rules.	10	L2	5	1
b)	Give a brief explanation on the types of VLSI testing depending on specific purpose it accomplishes.	10	L2	5	1